

Choosing and Applying a Convergence Test

Answer Key

AP Calculus BC

Quick Answers

- | | |
|---|--|
| 1. the limit-comparison ratio = 1, — | 7. the improper integral = $\frac{1}{\ln(2)}$, — |
| 2. the improper integral = $\frac{1}{e^2}$, — | 8. the exact sum = $\frac{\pi}{4}$, the error bound after five terms = $\frac{1}{11}$ |
| 3. the ratio limit = 0, the exact sum = e^2 | 9. the ratio limit = 0, — |
| 4. the limit of the term sizes = 0, the error bound after 100 terms = $\frac{1}{101}$, — | 10. the limit-comparison ratio = 1, the comparison series sum = $\frac{\pi^2}{6}$, — |
| 5. the limit-comparison ratio = 3, — | |
| 6. the ratio limit = ∞ , — | |
-

What is verified

14 of 22 answers machine-checked against SymPy, across 10 problems.

— marks an answer only you can judge — problems 1, 2, 4, 5, 6, 7, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus BC

LIM-7 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–5 of 5 across 22 tagged checks.

1. $\sum \frac{1}{n^2+3}$ by limit comparison with $\sum \frac{1}{n^2}$. (a) The limit. (b) The conclusion.

Solution (a): The quotient simplifies before any limit is taken:

$$\frac{1/(n^2+3)}{1/n^2} = \frac{n^2}{n^2+3} \rightarrow \boxed{1}$$

Solution (b) — instructor-judged. The limit is finite and *positive*, so the two series converge or diverge together. The comparison series is the p -series with $p = 2 > 1$, which converges, so $\sum \frac{1}{n^2+3}$ converges. Positivity matters because a limit of 0 or ∞ gives only a one-way conclusion. Full credit says that.

2. $\sum ne^{-n^2}$ by the integral test. (a) Evaluate $\int_1^\infty xe^{-x^2} dx$. (b) Conditions and conclusion.

Solution (a): With $u = x^2$, $du = 2x dx$:

$$\int_1^\infty xe^{-x^2} dx = \left[-\frac{1}{2}e^{-x^2} \right]_1^\infty = 0 + \frac{1}{2}e^{-1}.$$

So the integral equals

$$\boxed{\frac{1}{2e} \approx 0.184}$$

Solution (b) — instructor-judged. On $[1, \infty)$ the function $f(x) = xe^{-x^2}$ is positive, continuous, and decreasing (its derivative $e^{-x^2}(1 - 2x^2)$ is negative for $x \geq 1$). All three conditions hold, the integral is finite, so the series converges. The integral is *not* the sum — it bounds the tail, and the series sum lies between the integral and the integral plus the first term. Full credit gives both halves.

3. $\sum_{n \geq 0} \frac{2^n}{n!}$. (a) Limit of $\frac{2}{n+1}$. (b) The exact sum.

Solution (a): The denominator grows without bound while the numerator is fixed, so the ratio limit is

$$\boxed{0}$$

Since $0 < 1$, the ratio test gives absolute convergence.

Solution (b): This is the Maclaurin series for e^x evaluated at $x = 2$, so the sum is

$$\boxed{e^2 \approx 7.389}$$

4. Alternating harmonic series. (a) Limit of $1/n$. (b) Error bound after 100 terms. (c) Justify.

Solution (a): The term sizes are $\frac{1}{n}$, whose limit is

$$\boxed{0}$$

Solution (b): The remainder after N terms is at most the first omitted term. Using $n = 1$ through $n = 100$, the first omitted term has size $\frac{1}{101}$.

$$\boxed{\frac{1}{101} \approx 0.0099}$$

Solution (c) — instructor-judged. The alternating series test needs two things: the term sizes $\frac{1}{n}$ decrease, and their limit is 0. Both hold, so the series converges. The remainder bound then says $|S - S_N| \leq a_{N+1}$, which with $N = 100$ gives $\frac{1}{101}$. Full credit states the bound in that form rather than guessing a term count.

5. $\sum \frac{3n+1}{n^3+5}$ by limit comparison with $\sum \frac{1}{n^2}$. (a) The limit. (b) Conclusion and choice.

Solution (a):

$$\frac{(3n+1)/(n^3+5)}{1/n^2} = \frac{3n^3+n^2}{n^3+5} \rightarrow \boxed{3}$$

Solution (b) — instructor-judged. The limit is finite and positive, and $\sum \frac{1}{n^2}$ converges ($p = 2$), so the given series converges. The comparison series was chosen by keeping only the leading power top and bottom: $\frac{3n}{n^3} = \frac{3}{n^2}$, and the constant multiple does not affect convergence. Full credit explains that choice.

6. $\sum \frac{n!}{3^n}$. (a) Limit of $\frac{n+1}{3}$. (b) Conclusion, and what the terms show.

Solution (a): The numerator grows without bound while the denominator is constant, so the ratio limit is

$$\boxed{\infty}$$

Solution (b) — instructor-judged. A ratio limit greater than 1 (here infinite) means the series diverges. Independently, the terms themselves grow without bound: factorial growth outruns any fixed exponential, so $\frac{n!}{3^n} \rightarrow \infty$ and the n th-term test also gives divergence. Full credit gives both reasons.

7. $\sum_{n \geq 2} \frac{1}{n(\ln n)^2}$ by the integral test. (a) The improper integral. (b) Substitution, conditions, conclusion.

Solution (a): With $u = \ln x$, $du = \frac{dx}{x}$, the integral becomes $\int_{\ln 2}^{\infty} u^{-2} du = [-u^{-1}]_{\ln 2}^{\infty}$, so

$$\int_2^{\infty} \frac{dx}{x(\ln x)^2} = \boxed{\frac{1}{\ln 2} \approx 1.443}$$

Solution (b) — instructor-judged. The substitution is $u = \ln x$. On $[2, \infty)$ the integrand is positive, continuous and decreasing, so the integral test applies; the integral is finite, so the series converges. Full credit names the substitution and lists the three conditions.

8. $\sum_{n \geq 0} \frac{(-1)^n}{2n+1}$. (a) The exact sum. (b) Remainder bound after five terms.

Solution (a): This is the Leibniz series, the arctangent series at $x = 1$:

$$\boxed{\frac{\pi}{4}}$$

Solution (b): The first five terms are $n = 0$ through $n = 4$, so the first omitted term is the one with $n = 5$, of size $\frac{1}{2(5)+1}$.

$$\boxed{\frac{1}{11} \approx 0.0909}$$

9. $\sum \frac{n^2 3^n}{n!}$. (a) Limit of $\frac{3(n+1)}{n^2}$. (b) The cancellation and the conclusion.

Solution (a): The denominator grows faster than the numerator, so the ratio limit is

$$\boxed{0}$$

Solution (b) — instructor-judged. The ratio of consecutive terms is

$$\frac{(n+1)^2 3^{n+1}}{(n+1)!} \cdot \frac{n!}{n^2 3^n} = \frac{(n+1)^2}{n^2} \cdot 3 \cdot \frac{n!}{(n+1)!} = \frac{3(n+1)^2}{n^2(n+1)} = \frac{3(n+1)}{n^2}.$$

The factorial contributes the single factor $(n+1)$ in the denominator and the power of three contributes one factor of 3. Since the limit is $0 < 1$, the series converges absolutely. Full credit shows that cancellation.

10. $\sum \frac{n+2}{n^3-n+4}$; the ratio test returns 1. (a) Limit-comparison ratio. (b) Sum of $\sum 1/n^2$. (c) Explain.

Solution (a):

$$\frac{n+2}{n^3-n+4} \cdot n^2 = \frac{n^3+2n^2}{n^3-n+4} \rightarrow \boxed{1}$$

Solution (b): The comparison series is the p -series with $p = 2$, whose sum is

$$\boxed{\frac{\pi^2}{6}}$$

Solution (c) — instructor-judged. For terms that are a ratio of polynomials, consecutive terms differ by a factor that tends to 1, so the ratio test is guaranteed to return 1 and is guaranteed to be inconclusive — it is the wrong tool for rational terms, not merely unlucky. Limit comparison against $\frac{1}{n^2}$ gives a finite positive limit, and since that p -series converges, so does the given series. Full credit explains why the ratio test had to fail.